# Module 10: Epigenetics — Study Questions

1. In Module 07 you learned: DNA → RNA → Protein. What does it mean for a gene to be "expressed"?

2. Every cell in your body got the same DNA through mitosis (Module 08). So why does a muscle cell look completely different from a nerve cell?

3. What is gene regulation? Why do cells need it?

4. What is a transcription factor? How does it relate to the promoter you learned about in Module 07?

5. In your own words, what is epigenetics? How is it **different** from a mutation?

6. What is DNA methylation? Does it turn a gene on or off? (Use the memory trick: **M**ethylation = _____)

7. What are histones? What happens when histones are acetylated? (Use the memory trick: **A**cetylation = _____)

8. What is X-inactivation? What is a Barr body?

9. How do calico cats show epigenetics in action? Why are almost all calico cats female?

10. Give two examples of how the environment can change which genes are turned on or off. (Connect this to Module 09: phenotype = genotype + environment.)

11. Identical twins share the same DNA. Using what you know about epigenetics, why might they develop different traits or health conditions as adults?

12. **Putting it all together:** A liver cell and an eye cell have the same DNA (Module 08 — mitosis), the same genes (Module 07), and the same alleles (Module 09). Using what you learned in this module, explain how they end up looking and functioning so differently.